

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (A)

Gandipet, Hyderabad -75

B. E (ECE) Assignment-1, November 2024

SEMESTER: I

Section: ECE-1

SUBJECT: PROBLEM SOLVING AND PROGRAMMING USING PYTHON(22CSC40)

Date: 05/11/2024

Max. Marks: 10

Answer ALL questions. CO: Course Outcomes, BT: Blooms Taxonomy

SET-1(Roll No. 1 to 16)

1. What is an algorithm, and why is it important in problem-solving? [CO1] [BT1] [2M]
2. Discuss the role of the compiler, linker, and loader in the software development process. [CO1] [BT6] [3M]
3. Develop a Python program to convert temperature from Fahrenheit to Celsius and vice versa ($c=(f-32)/1.8$ and $f=(c*1.8)+32$). [CO2] [BT6] [3M]
4. Develop a program to create a tuple of numbers and find the maximum and minimum values. [CO2] [BT6] [2M]

SET-2(Roll No. 17 to 32)

5. Demonstrate the advantages and disadvantages of Flowcharts. [CO1] [BT2] [2M]
6. Develop an algorithm for swapping two numbers WITHOUT USING extra variable and create a flowchart to visualize your solution. [CO1] [BT6] [3M]
7. Compare a list and a tuple in Python. [CO2][BT2] [2M]
8. The displacement of a body (s) against time (t) where (u) is the initial velocity and (a) is the acceleration. Develop a Python program to find S for given u, t, a. [CO2][BT6] [3M]

SET-3(Roll No. 33 to 48)

9. Compare the top-down and bottom-up programming methodologies. [CO1] [BT2] [2M]
10. Develop an algorithm for swapping two numbers USING extra variable and create a flowchart to visualize your solution. [CO1] [BT6] [3M]
11. Develop a Python program to create a list, add a new item, and remove an existing item. [CO2] [BT6] [3M]
12. Develop a Python program to count number of characters in a given string. [CO2] [BT6] [2M]

SET-4(Roll No. 49 to 64)

13. Compare the Interactive Mode and Script Mode of Python Interpreters. **[CO1] [BT2] [2M]**
14. Design a small Python script that takes user input and utilizes identifiers and literals, incorporating comments to explain your code. **[CO2] [BT6] [3M]**
15. Develop a Python program to create a dictionary to store student names and their grades, and then print each student's name with their grade. **[CO2] [BT6] [3M]**
16. Develop a Python program to find average of a given list. **[CO2] [BT6] [2M]**

Note: Submit the Assignment-2 hard copy by 12/11/2024.